

Real-Time Conversational Query Expansion & Hierarchical Topic Classification — System Approach

1. User sends a message through the Streamlit chat interface.
2. The Conversation Manager stores both user and bot messages in a fixed sliding window containing the latest 20 messages.
3. The Context Builder fetches the sliding window history and creates a structured conversational context prompt.
4. The Query Expander LLM uses the current query plus conversation context to resolve pronouns, ambiguity, and incomplete sentences into a semantically complete query.
5. The expanded query is then passed to the Topic Hierarchy Engine instead of the raw user query to improve classification accuracy.
6. The Level 1 classifier first predicts the top-level domain such as Politics, Finance, Sports, Tech, Health, or General.
7. Based on the Level 1 prediction, the query is routed to the corresponding Level 2 domain-specific classifier.
8. The Level 2 classifier predicts the fine-grained subdomain such as India, UK, Cricket, AI, Banking, etc.
9. Confidence thresholds are applied to detect uncertain predictions and trigger fallback handling or clarification when confidence is low.
10. The Response Aggregator combines the original query, expanded query, topic labels, and confidence scores into a unified inference output.
11. The final output is displayed in real time on the Streamlit UI along with the expanded query and hierarchical topic prediction.

Alternative Approaches Evaluated

Approach	Why Rejected
Pure LLM Classification	Inconsistent outputs, hallucinations, higher latency
Multiple LLM Calls	Better accuracy possible, but not production-grade due to cost and latency
Single Flat Classifier	Poor scalability with increasing topic combinations
Parallel Pipeline	Weak for incomplete conversational queries

Final Design Choice

- LLM used only once for contextual query expansion
- Fine-tuned supervised models used for hierarchical topic classification
- Reduces latency, improves consistency, and is more production-oriented

Planned Verticals & Sub domains

Vertical	Sub-domain
Politics	Indian Politics
	UK Politics
	USA Politics
	International Relations
	Elections & Voting
	Government Policy
	Political Parties
	Defence & Military
	Geopolitics
	Judiciary & Law
Sports	Cricket
	Football (Soccer)
	Tennis
	Athletics & Olympics
	Basketball
	Kabaddi & Wrestling
	Badminton
	Formula 1 & Motorsport
	Hockey
	Sports Business
Technology	Artificial Intelligence
	Smartphones & Gadgets
	Space & Exploration

Vertical	Sub-domain
	Cybersecurity
	Software & Apps
	Social Media & Internet
	Electric Vehicles & Clean Tech
	Big Tech Companies
	Semiconductors & Chips
	Telecom & Networks
Health / Medicine	Diseases & Conditions
	Mental Health
	Drugs & Medications
	Nutrition & Diet
	Fitness & Exercise
	Public Health & Epidemics
	Surgery & Procedures
	Maternal & Child Health
	Alternative Medicine
	Health Policy
Finance	Stock Market
	Cryptocurrency
	Banking & Loans
	Personal Finance
	Mutual Funds & SIP
	Insurance
	Real Estate
	Taxation
	Global Economy

Vertical

Sub-domain

Startups & VC

General

Greetings

Chitchat

Interruptions

Acknowledgements

Out of Scope